

Savitzky-Golay optimizer diagnostics

Objective: mean squared translational/angular acceleration residual; angular residual uses the reference inertia/mass metric

Tolerances: {'ftol': None, 'xtol': None, 'gtol': 1e-06, 'note': 'ftol and xtol are disabled by default in the SG estimator; gtol or max_nfev terminates the solve unless explicitly overridden'}

Selected strict-ZOH solve

status=1 success=True nfev=4 njev=4

message='gtol' termination condition is satisfied.

reported optimality=2.7666576585261282e-08

```
initial={'cost': 1.5031743504563846, 'gradient_inf_norm': 0.03377991928460311,
'gradient_l2_norm': 0.04203309160534639, 'jacobian': {'condition_number_nonzero_subspace':
410.10909906722264, 'numerical_rank': 14, 'numerical_tolerance': 8.60331345037508e-11,
'singular_values': [119.51227405694257, 118.47088480712405, 102.44000579293615,
48.62808732527297, 44.72150752042473, 25.85992473866219, 12.980547791209727,
11.713724019823204, 11.32871201195832, 11.045134784111513, 9.859077652330727,
3.7972744719837994, 3.7925785549485496, 0.29141580698591824]}, 'residual_l2_norm':
1.733882551072237}
final={'cost': 1.5031742302822693, 'gradient_inf_norm': 2.7666576585261282e-08,
'gradient_l2_norm': 4.589470844325769e-08, 'jacobian': {'condition_number_nonzero_subspace':
410.1436198608981, 'numerical_rank': 14, 'numerical_tolerance': 8.603299881304802e-11,
'singular_values': [119.51208556323547, 118.47075033231992, 102.43999338997295,
48.62811327321679, 44.72150686538878, 25.859926873382303, 12.980541940829838,
11.713663420674319, 11.32865033705389, 11.045094983031847, 9.859029892507493,
3.797276519134773, 3.7925765961610374, 0.29139081964451496]}, 'residual_l2_norm':
1.7338824817629765}
coordinate_step_l2_norm=5.11637436405372e-05
cost_reduction=1.2017411532561084e-07
gradient_tolerance_satisfied=True
finite_difference_max_relative_error=9.027424324664627e-12
finite_difference_directional_checks=[{'analytic_l2_norm': 96.8186314818096,
'difference_l2_norm': 8.740228689218589e-10, 'direction': 'negative_gradient',
'finite_difference_l2_norm': 96.81863148205667, 'relative_error': 9.027424324664627e-12,
'scheme': 'central', 'step': 6.0554544523933395e-06}]
```

All smoothstep and strict-ZOH stage details are in optimizer_diagnostics.json.